IPL Vision — Project Introduction

IPL Vision

Cricket Analytics and Match Win Prediction System

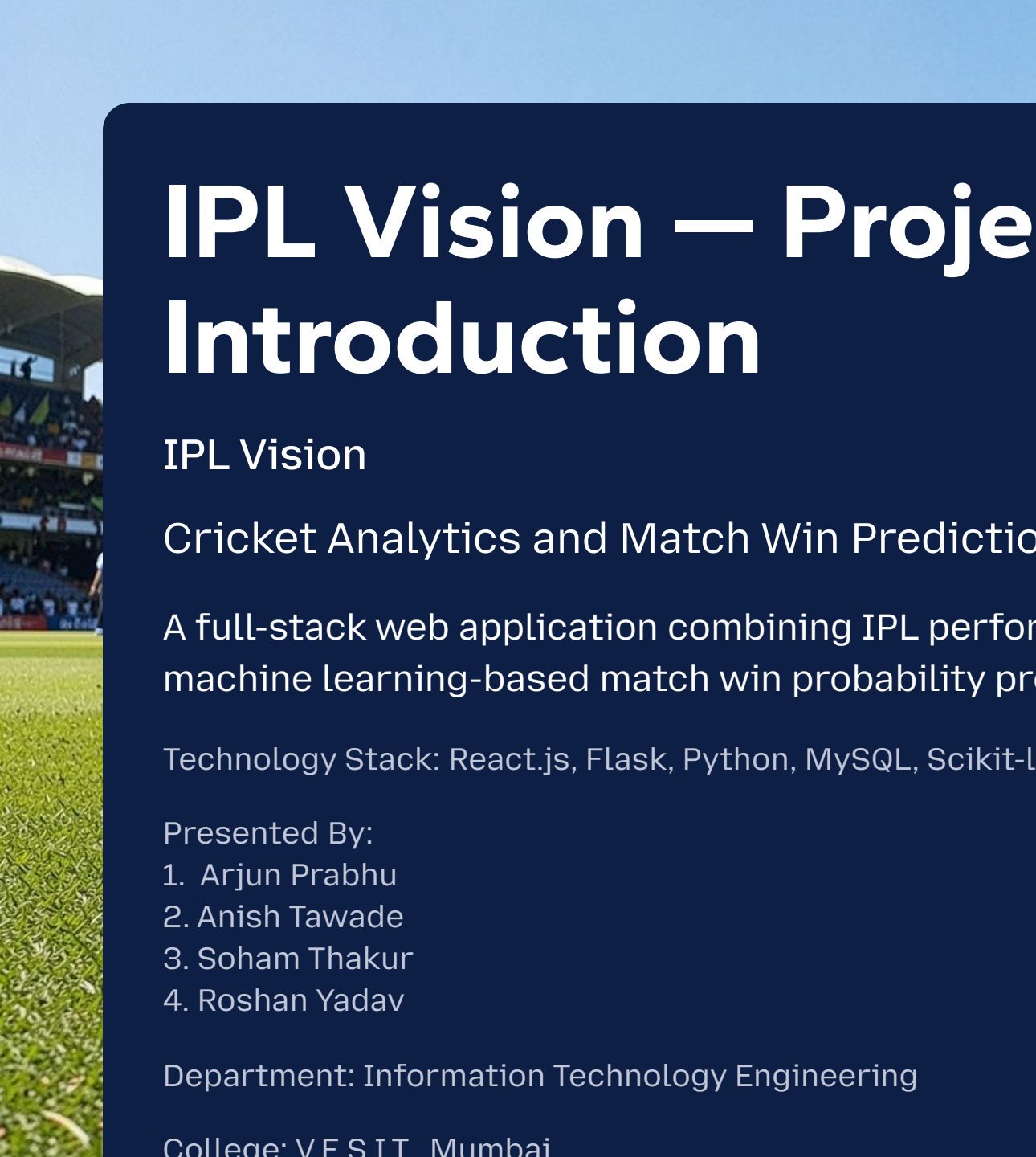
A full-stack web application combining IPL performance analytics with machine learning-based match win probability prediction.

Technology Stack: React.js, Flask, Python, MySQL, Scikit-learn

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Problem Statement and Objectives

Problem Statement

Analyzing multiple IPL seasons manually makes it difficult to identify player performance trends, compare team statistics, and estimate winning probabilities as match situations change.

IPL Vision integrates historical cricket analytics and machine learning-based win prediction into a single interactive platform.

Objectives

- Analyze batter and bowler performance across IPL seasons.
- Provide team-level statistics and comparisons.
- Predict match-winning probabilities during second-innings chases.
- Integrate data analytics and machine learning into a web application.
- Improve backend efficiency through centralized data caching.

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System Architecture and Technology Stack

01

Data Layer

- Historical IPL datasets: deliveries.csv and matches.csv.
- Centralized data loader for preprocessing and in-memory caching.

02

Backend — Python and Flask

- REST APIs for batter, bowler, and team analytics.
- Match win prediction API.
- Statistical computations using Pandas and NumPy.

03

Machine Learning

- Scikit-learn pipeline.
- OneHotEncoder and StandardScaler.
- Calibrated LinearSVC classifier.

04

Frontend — React.js

- Interactive analytics dashboards.
- Chart.js visualizations.
- Match prediction interface with probability bars.

Technology Stack

React 19 | Vite 6 | Tailwind CSS 4 | Flask 2.2 | Pandas | NumPy | Scikit-learn | Chart.js | Zustand



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Core Modules and Features

1. Batter Analysis

- Career and season-wise batting statistics.
- Runs, strike rates, boundary distributions, and dismissal analysis.

2. Bowler Analysis

- Bowling economy and strike rates.
- Powerplay, middle-over, and death-over performance.
- Opponent and batter matchup analysis.

3. Team Analytics

- Franchise win-loss records.
- Team performance comparisons.
- Titles and top-performing players.

4. Match Win Predictor

- Accepts teams, venue, target, current score, overs, and wickets lost.
- Calculates runs left, current run rate, and required run rate.
- Displays estimated winning probabilities for the chasing and defending teams.



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Machine Learning Model and Dataset

Dataset

- Historical IPL data from 2008–2024.
- More than 1,095 matches.
- Over 260,000 ball-by-ball delivery records.

Model

Calibrated Linear Support Vector Machine (LinearSVC)

- Platt probability calibration using 3-fold cross-validation.
- Binary classification of match outcomes.

Input Features

- Batting team and bowling team.
- City or venue.
- Runs left and balls left.
- Wickets left and target runs.
- Current Run Rate (CRR).
- Required Run Rate (RRR).

Training and Testing

- Total game states: 103,833
- Training set: 83,066 (80%)
- Testing set: 20,767 (20%)

01 Data Preprocessing

02 Feature Engineering

03 Encoding and Scaling

04 Model Training

05 Probability Calibration

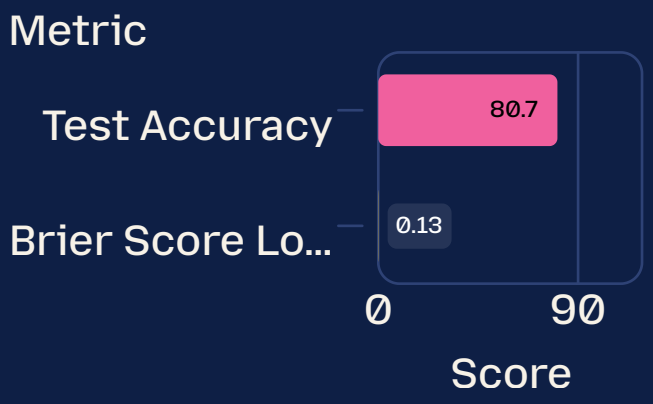
06 Evaluation

Pipeline: Data Preprocessing → Feature Engineering → Encoding and Scaling → Model Training → Probability Calibration → Evaluation

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Model Evaluation and Accuracy Results

Overall Model Performance



Class-wise Classification Results

Class	Precision	Recall	F1-Score
Defending Team Wins	0.81	0.78	0.80
Chasing Team Wins	0.80	0.83	0.82
Macro Average	0.81	0.81	0.81
Weighted Average	0.81	0.81	0.81

Evaluation Summary

- Achieved 80.69% accuracy on 20,767 held-out test examples.
- Achieved a macro F1-score of 0.81.
- Obtained a Brier score of 0.1331 for probability prediction.

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Thank You & Q&A

Thank You!

IPL Vision

Cricket Analytics and Match Win Prediction

A unified platform for IPL performance analysis and machine learning-based match prediction.

Questions & Answers

Thank you for your time and attention.

Presented By: [Team Members' Names]

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